Customer Churn Analysis

Executive Summary

This project focuses on analyzing customer churn to identify key factors driving customer attention and to support data-driven retention strategies. Using structured customer data, analysis was performed to uncover patterns related to contract type, tenure, monthly charges, and service usage.

The findings reveal that customers with month-to-month contracts, shorter tenure, and higher monthly charges are more likely to churn. Additionally, the absence of value-added services such as premium technical support contributes significantly to higher churn rates.

Based on these insights, the project highlights opportunities for improving customer retention through targeted strategies such as promoting long-term contracts, enhancing customer support services, and focusing on high-risk customer segments.

This analysis demonstrates how data analytics and visualization tools can be effectively used to generate actionable business insights and improve overall customer retention performance.

Business Problem

• The company is experiencing high customer churn, leading to revenue loss and increased acquisition costs.

• There is a lack of clear understanding of why customers are leaving and which segments are most at risk.

• Key factors influencing churn—such as contract type, tenure, pricing, and service usage—are not being effectively analyzed.

• Without data-driven insights, the business is unable to implement targeted retention strategies and improve customer lifetime value.

SQL Data Preparation & Churn Analysis

The dataset was processed and analyzed using SQL to perform data cleaning, feature engineering, and in-depth churn analysis. The goal was to identify key factors influencing customer churn and revenue loss.

Data Understanding & Validation

Initial queries were executed to understand the dataset structure and ensure data quality:

- Checked total number of records and column structure
- Identified missing values in key columns such as *total_charges* and *monthly_charge*
- Verified data integrity by checking for duplicate customer records

Data Cleaning

- Missing values were identified and handled appropriately to ensure accurate analysis
- Duplicate records were checked using customer_id to maintain uniqueness
- Data consistency was ensured across all fields before further processing

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Feature Engineering

Several new features were created to enhance analysis:

- **Churn Flag:**
  A binary column (churn_flag) was created to classify customers as churned (1) or retained (0).
- **Tenure Grouping:**
  Customers were segmented into tenure-based groups:
  - 0–6 months
  - 6–12 months
  - 1–2 years
  - 2+ years
- **Average Revenue per Month:**
  Calculated by dividing total revenue by tenure, providing a normalized measure of customer value.
- **Population Enrichment:**
  Customer data was joined with external population data to analyze churn based on geographic density.

Churn Analysis

- **Overall Churn Rate:**
  The percentage of customers who discontinued services was calculated to understand overall retention performance.
- **Churn by Contract Type:**
  Customers with short-term or flexible contracts showed higher churn rates compared to long-term contracts.
- **Churn by Payment Method:**
  Certain payment methods were associated with higher churn, indicating possible friction in billing or customer experience.
- **Churn by Tenure:**
  New customers (0–6 months) exhibited significantly higher churn rates, highlighting early-stage customer risk.

- **Churn by Internet Service:**
  Customers with specific internet types or no internet service showed varying churn behaviors.
- **Behavioral Analysis:**
  Service usage features such as streaming services were analyzed to identify patterns in churn behavior.

Revenue Impact Analysis

- Total revenue and revenue lost due to churn were calculated
- High churn segments were identified as key contributors to revenue loss
- Top customers were ranked based on revenue contribution

Customer Segmentation

- Customers were segmented into revenue quartiles using window functions
- Churn rates were compared across these segments
- Results indicated that lower-revenue segments had higher churn rates

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Churn Drivers

- Churn categories and reasons were analyzed to identify primary causes
- Top reasons included service dissatisfaction, pricing issues, and competition
- Population-based segmentation revealed differences in churn across high, medium, and low-density areas

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Advanced Analysis

- Multi-factor analysis combining contract type, internet service, and technical support was performed
- This helped identify high-risk customer profiles with the highest churn probability

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Insights

- Maven has 27% of churn rate and only 6% are joining.
- 42% of churned customers only stayed for 6 months or less.
- The top 3 reasons for churn are competitors made better offers, competitors had better devices and attitude of support staff.

- Maven lost ~$1.7 million to competitors, making it the most expensive type of churn
- The key indicators of churn are Month-to-Month contract , No Premium Tech Support, Fiber Optic internet, No promotional offer and Offer E.
- 70% of customers who churned to competitors used Fiber Optic
- High value customers are churning at a rate of 16%

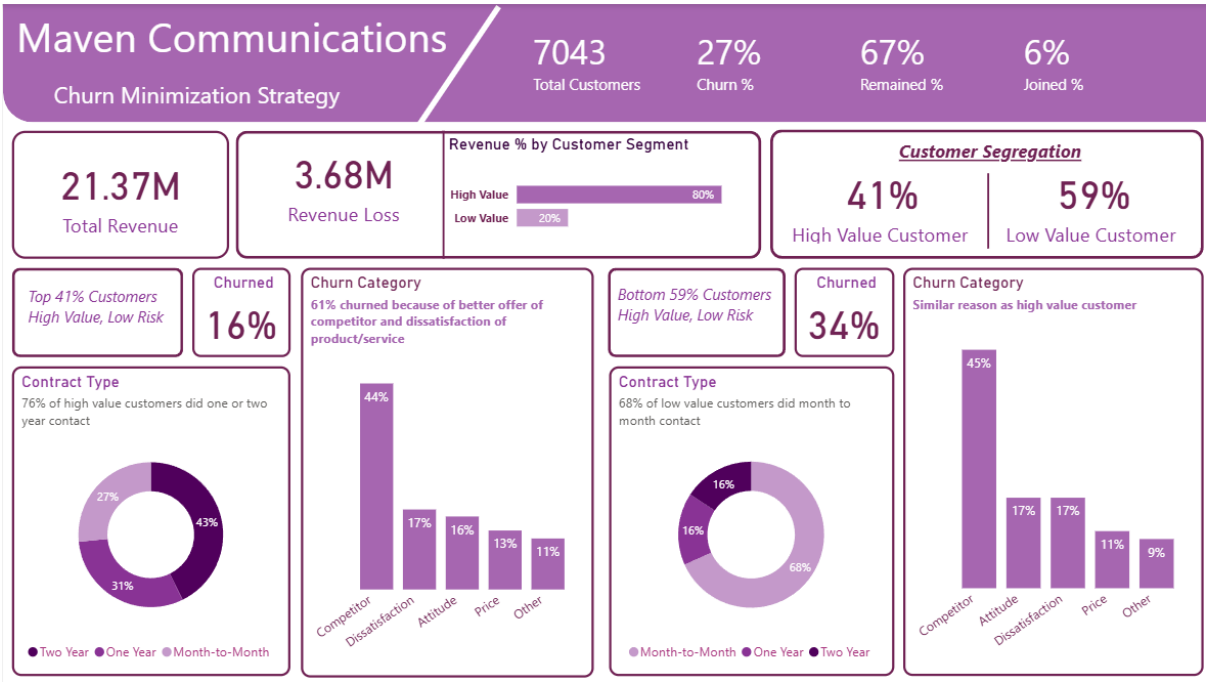
Power BI Dashboard Building

An interactive Power BI dashboard was developed to visualize key metrics and support data-driven decision-making. The dashboard provides insights into customer churn patterns, revenue trends, and high-risk customer segments.

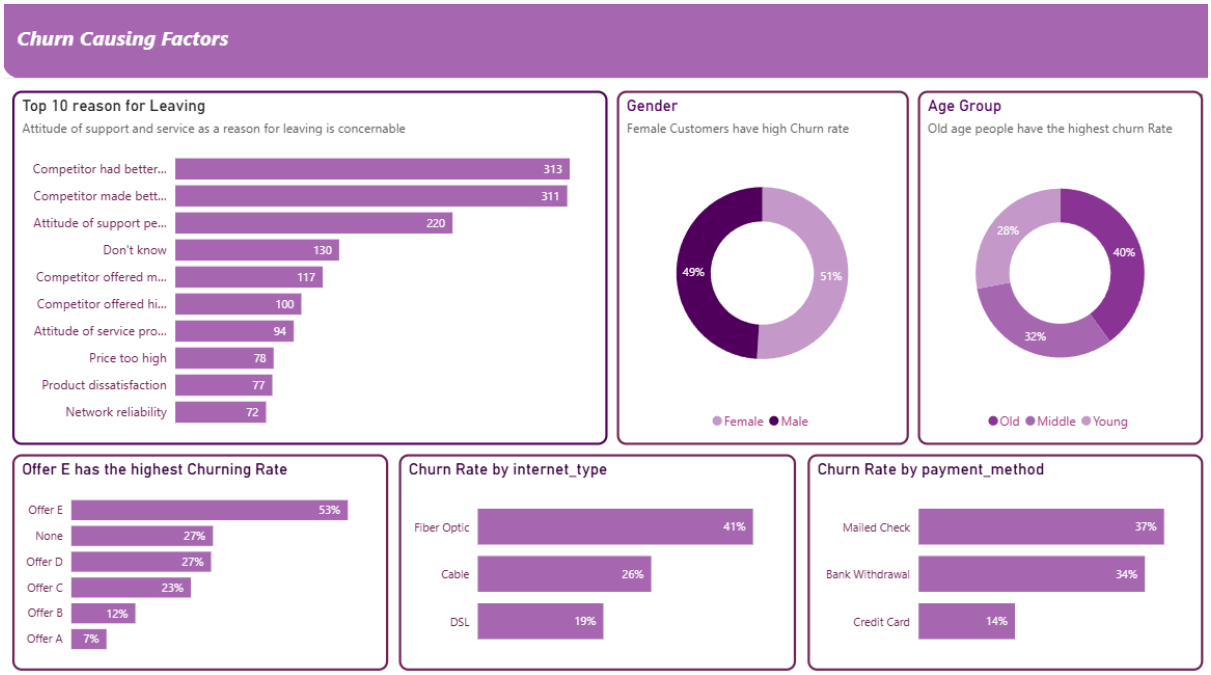
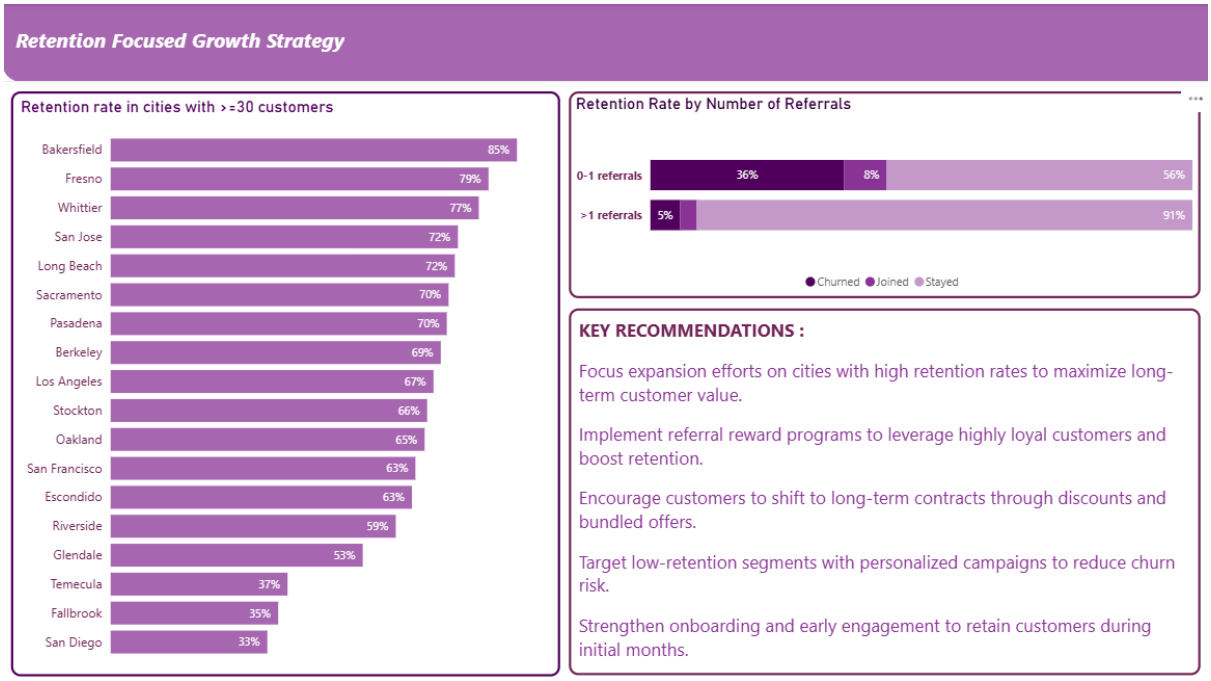
Key visuals included:
• Churn rate by contract type
• Customer distribution by tenure
• Monthly charges vs churn analysis
• High-risk customer segmentation

These visualizations enable stakeholders to quickly identify problem areas and take targeted actions to improve customer retention.

The Dashboard page displays KPIs for the Total Revenue, Total Customers, Churn Rate .I have segmented the customers into high value customers and low value customers and the find out the difference factors affecting the churn rate on these two segments, and also, find the revenue contribution of both the segments.



The Churn Causing Factors page shows the top 10 reasons for leaving and how the internet type, payment method and offers are affecting the churn rate and also find out which age group and gender have highest churn rate.



The Retention Focused Growth Strategy Page shows the retention rate of cities with more than 30 customers and retention rate by no of referrals so that we can focus these cities and customers to enhance our profit margin and boost retention.